Robotic-Lawnmower Algorithm
Group C

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Goals for Day 1

- Getting started with the Simulink-model
- Complying with requirements:
    - No sensor coil should stay outside of the working area for more than three seconds
    - One coil has always to be inside
    - Coverage of more than 70%

- Not needed: return home function

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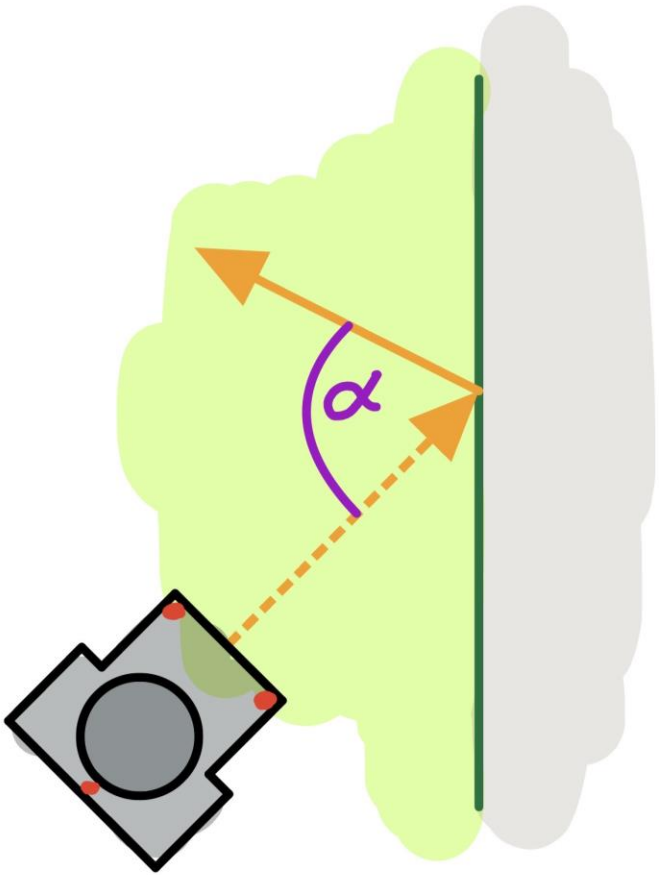
Tasks for Day 1

1. Implement state flow Chart
2. Ensure the mower can only operate within the garden
3. Maximise area coverage



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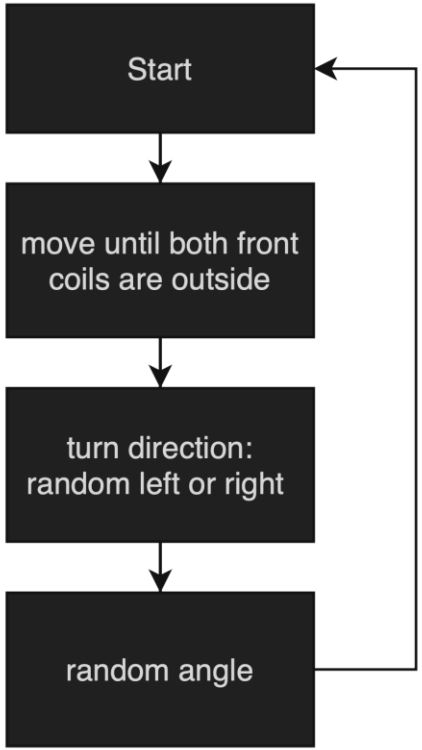
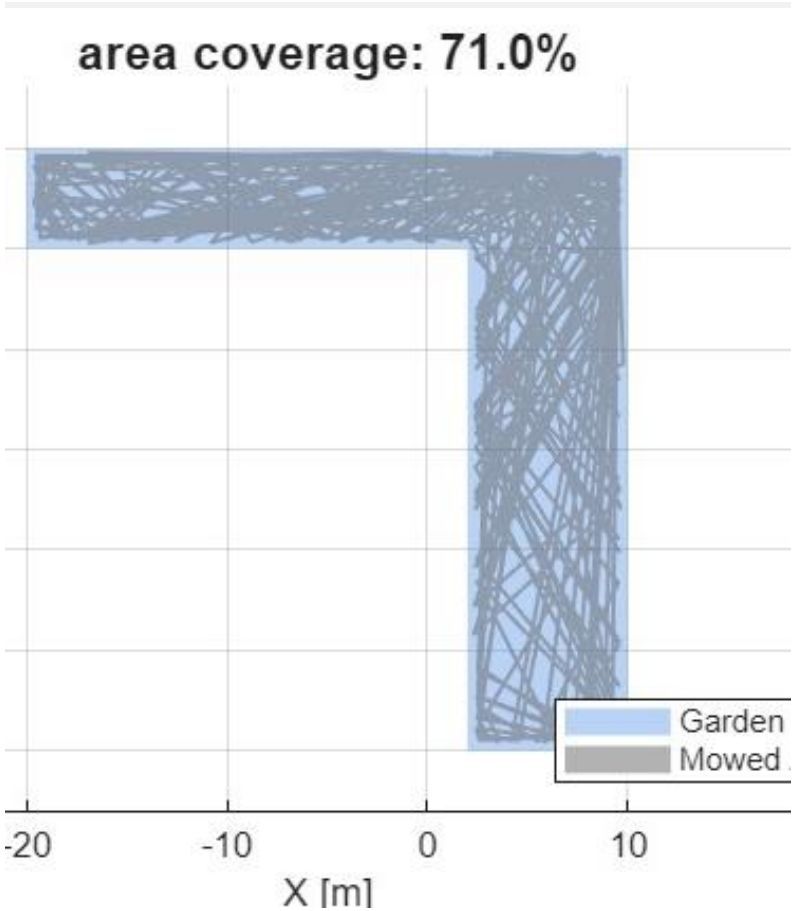
Random-Walk





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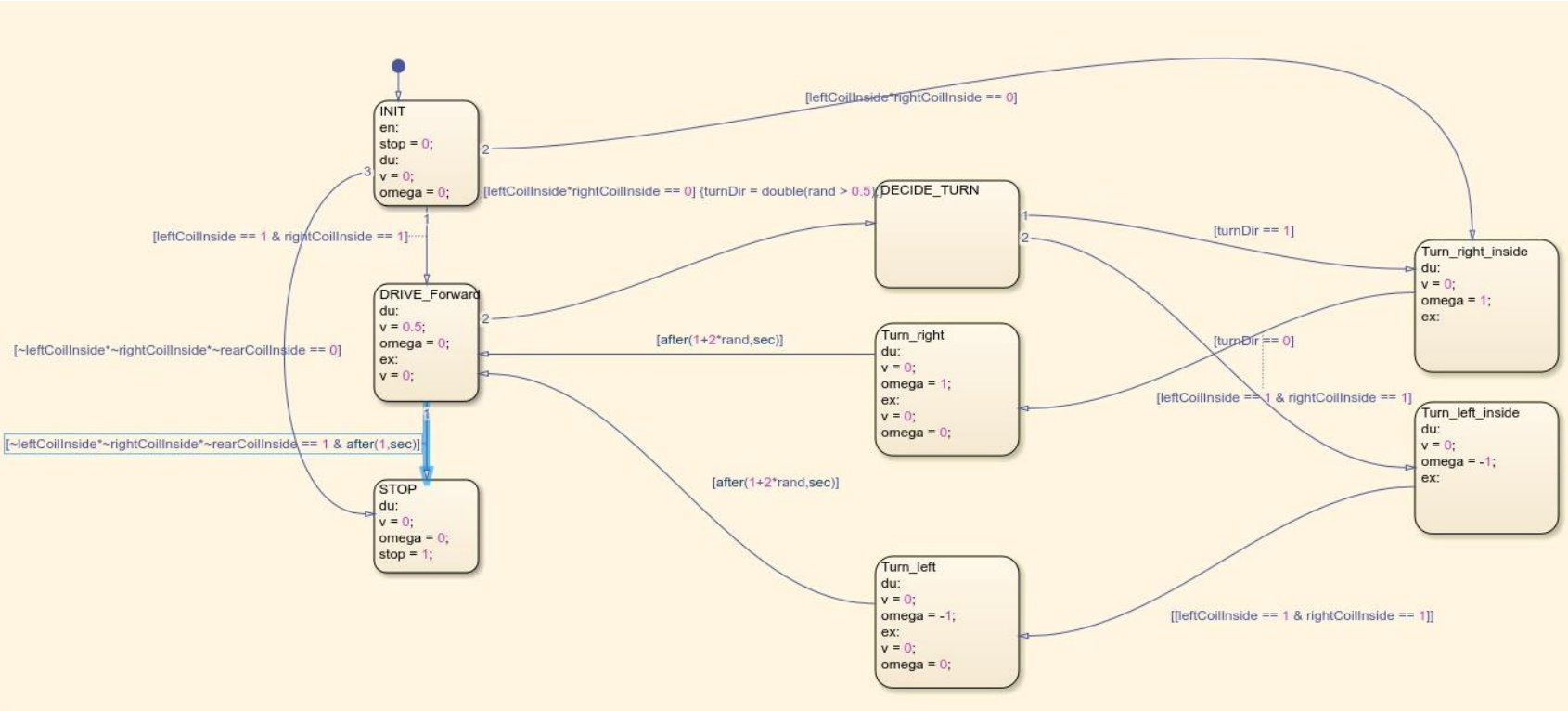
State Flow Chart



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Future Improvements

- Replace random turns with a more structured turning strategy (angle calculation)
- Use a systematic path-planning approach for simple garden shapes

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Advantages

- Easy to implement
- Reliable
- Low computational effort
- Low memory usage

Disadvantages

- Low efficiency
- Drives over the same spots frequently

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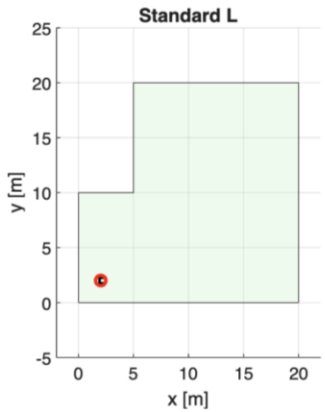
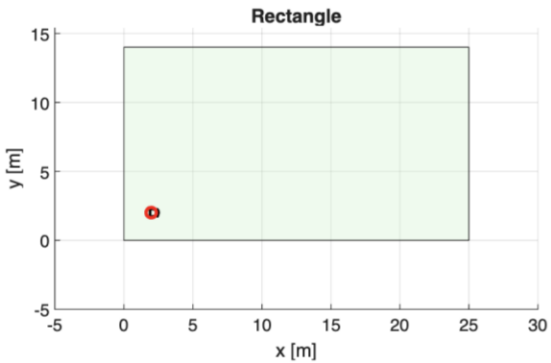
Tasks for Day 2

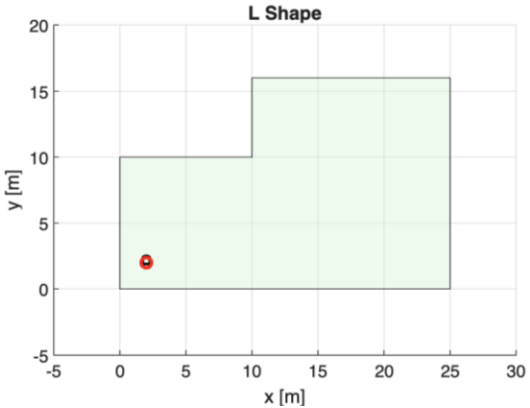
- Add UI (priority high)
  - Dropdown-menu for garden shape
  - Dropdown-menu for algorithm
  - Comparison of results
- Model (priority medium)
  - Parameter optimization: velocity, angle
- Algorithm (priority low)
  - If time left, optimize algorithm

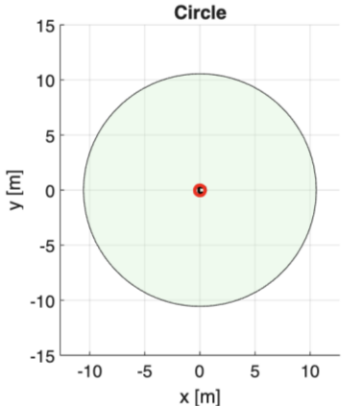
Alternative Garden Shapes

- New garden shapes
- All with the size: 350 m²









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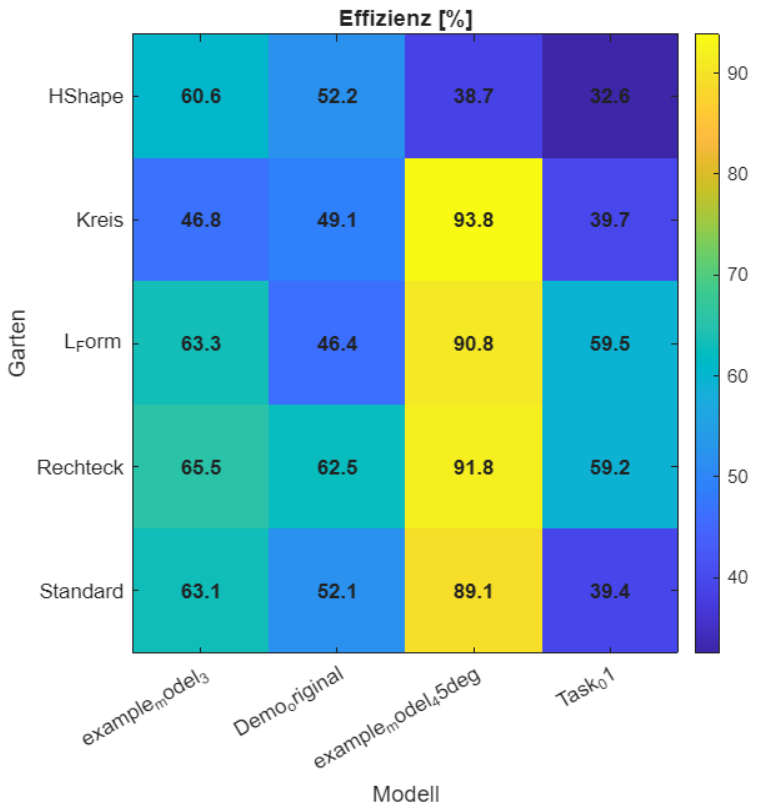
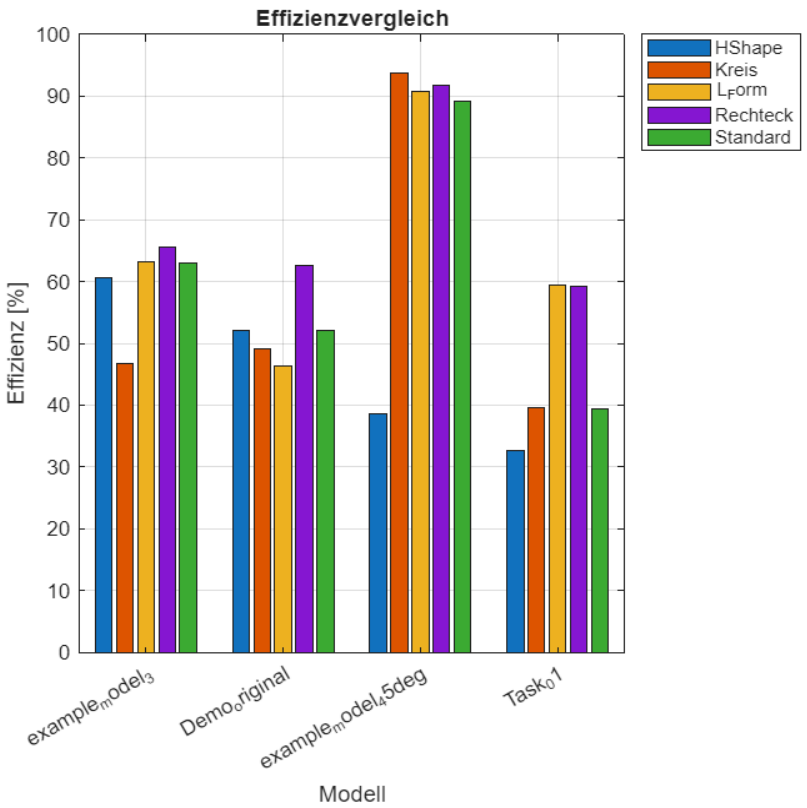
Comparison





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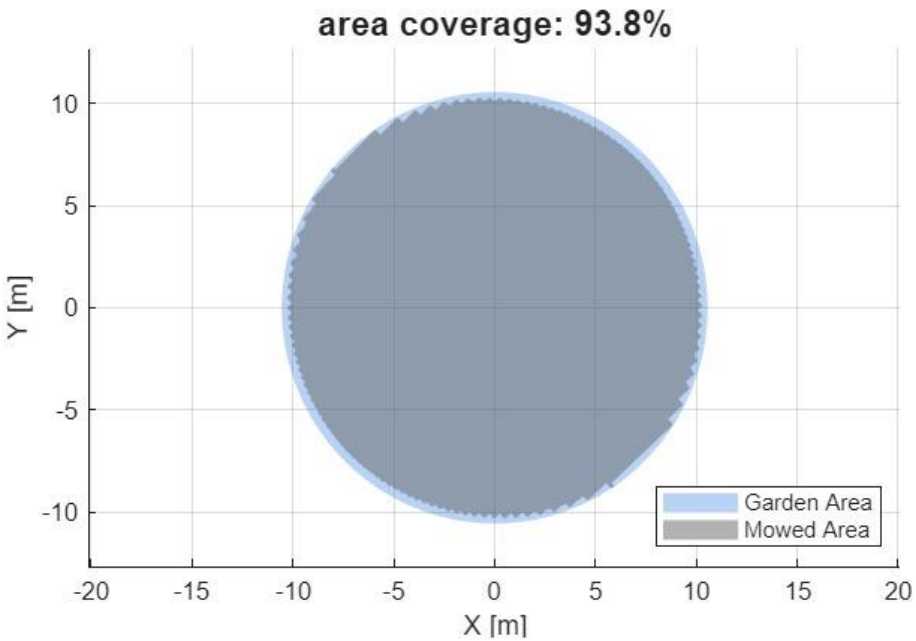
Impact of AI on Productivity

- Other ideas and approaches
- Faster progress
- Increased productivity
- "Vibe coding"
- Used Models: ChatGPT and Gemini
    - Garden shapes
    - Graphic design of figures
    - Explanation of code



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Thank you for your attention

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